

Permutation and Combination

1. In how many ways can the letters of the word 'SUNDAY' be arranged?

- A) 720
- B) 120
- C) 360
- D) 240

Answer: A) 720

2. Find the value of 8P_4 .

- A) 1680
- B) 1600
- C) 70
- D) 40320

Answer: A) 1680

3. In how many ways can a committee of 4 be chosen from 12 distinct people?

- A) 495
- B) 11880
- C) 48
- D) 480

Answer: A) 495

4. How many 3-letter words can be formed using the letters A, B, C, D, E without repetition?

- A) 60
- B) 125
- C) 15
- D) 20

Answer: A) 60

5. In how many ways can 7 friends stand in a straight line?

A) 5040

B) 720

C) 49

D) 343

Answer: A) 5040

6. Find the number of diagonals in an octagon (8-sided polygon).

A) 20

B) 28

C) 14

D) 12

Answer: A) 20

7. How many ways can 5 distinct keys be arranged in a circular key ring?

A) 12

B) 24

C) 60

D) 120

Answer: A) 12

8. In how many ways can the letters of the word 'GOOGLE' be arranged?

A) 180

B) 360

C) 720

D) 90

Answer: A) 180

9. How many 4-digit numbers can be formed using digits 1, 2, 3, 4, 5 if repetition is allowed?

A) 625

B) 120

C) 256

D) 1024

Answer: A) 625

10. From a group of 8 men and 7 women, 6 persons are to be selected to form a committee so that there are exactly 3 men and 3 women.

A) 1960

B) 2400

C) 11760

D) 560

Answer: A) 1960

11. Find the number of triangles that can be formed by joining the vertices of a decagon (10-sided polygon).

A) 120

B) 45

C) 100

D) 90

Answer: A) 120

12. In how many ways can 6 different books be arranged on a shelf such that two specific books are always together?

A) 240

B) 120

C) 720

D) 480

Answer: A) 240

13. How many integers greater than 3000 can be formed with the digits 1, 2, 3, 4, 5 using each digit at most once?

A) 120

B) 72

C) 192

D) 240

Answer: C) 192

14. How many ways can a team of 5 be selected from 10 players if the captain must always be included?

A) 126

B) 252

C) 210

D) 120

Answer: A) 126

15. In how many ways can the letters of the word 'MACHINE' be arranged so that the vowels may occupy only odd positions?

A) 576

B) 144

C) 288

D) 720

Answer: A) 576

16. How many ways can 8 beads of different colors be strung into a necklace?

A) 2520

B) 5040

C) 40320

D) 20160

Answer: A) 2520

17. Find the number of arrangements of the letters of the word 'ALGEBRA' without changing the relative order of vowels and consonants.

A) 720

B) 120

C) 2520

D) 5040

Answer: B) 120

18. If $10C_r = 10C_{(r+2)}$, find the value of $5C_r$.

- A) 5
- B) 10
- C) 1
- D) 6

Answer: A) 5

19. A box contains 5 red, 4 blue, and 3 green balls. In how many ways can 3 balls be drawn such that at least 1 is red?

- A) 185
- B) 220
- C) 35
- D) 120

Answer: A) 185

20. How many 3-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5 if repetition is not allowed?

- A) 100
- B) 120
- C) 150
- D) 180

Answer: A) 100

21. In how many ways can 10 people shake hands with each other?

- A) 45
- B) 90
- C) 50
- D) 100

Answer: A) 45

22. A paper consists of 2 sections having 5 questions each. A student must answer 6 questions in all, selecting at least 2 from each section. In how many ways can this be done?

- A) 200
- B) 100
- C) 150
- D) 250

Answer: A) 200

23. How many words can be formed by rearranging the letters of the word 'ASCENT' such that A and T occupy the first and last positions respectively?

- A) 24
- B) 120
- C) 720
- D) 48

Answer: A) 24

24. If $nP_4 = 360$, find n .

- A) 6
- B) 5
- C) 7
- D) 8

Answer: A) 6

25. The number of ways in which 5 boys and 5 girls can be seated in a row so that no two girls are together is:

- A) 86400
- B) 14400
- C) 7200
- D) 28800

Answer: A) 86400

26. How many 5-digit odd numbers can be formed using digits 0, 1, 2, 3, 4 without repetition?

A) 48

B) 36

C) 72

D) 96

Answer: B) 36

27. Find the number of parallelograms that can be formed from a set of 5 parallel lines intersecting another set of 4 parallel lines.

A) 60

B) 50

C) 40

D) 20

Answer: A) 60

28. In how many ways can 3 letters be posted in 5 letterboxes?

A) 125

B) 243

C) 60

D) 15

Answer: A) 125

29. How many words can be formed using the letters of the word 'BANANA'?

A) 60

B) 120

C) 720

D) 360

Answer: A) 60

30. In how many ways can 9 different books be distributed among 3 students such that each receives 3 books?

A) 1680

B) 5040

C) 280

D) 1260

Answer: A) 1680

31. If $nC_{12} = nC_8$, find nC_2 .

A) 190

B) 20

C) 153

D) 45

Answer: A) 190

32. A bag contains 4 white, 5 red, and 6 blue balls. Three balls are drawn at random. In how many ways can they be all red?

A) 10

B) 20

C) 455

D) 15

Answer: A) 10

33. How many 4-digit numbers can be formed using the digits 1, 2, 3, 4, 5, 6 divisible by 4 (repetition not allowed)?

A) 60

B) 84

C) 96

D) 72

Answer: A) 60

34. How many signals can be made by hoisting 3 flags of different colors one above the other from 6 flags of different colors?

A) 120

B) 60

C) 20

D) 216

Answer: A) 120

35. In how many ways can a cricket team of 11 be selected from 14 players if 2 particular players are always included?

A) 220

B) 364

C) 66

D) 286

Answer: A) 220

36. Find the number of ways to arrange the letters of the word 'GARDEN' with the vowels in alphabetical order.

A) 360

B) 720

C) 120

D) 240

Answer: A) 360

37. How many distinct permutations can be made from the letters of the word 'MATTER'?

A) 360

B) 720

C) 120

D) 180

Answer: A) 360

38. There are 12 points in a plane, of which 5 are collinear. The number of straight lines that can be formed by joining these points is:

A) 57

B) 66

C) 56

D) 60

Answer: A) 57

39. A committee of 7 is to be chosen from 13 students. In how many ways can this be done?

A) 1716

B) 1287

C) 720

D) 5040

Answer: A) 1716

40. In how many ways can 7 people be seated in a row if two specific people must not sit together?

A) 3600

B) 5040

C) 1440

D) 4320

Answer: A) 3600

41. Find the rank of the word 'ZENITH' in a dictionary.

A) 616

B) 615

C) 614

D) 613

Answer: A) 616

42. How many numbers greater than 24000 can be formed using digits 1, 2, 3, 4, 5 without repetition?

A) 78

B) 120

C) 96

D) 72

Answer: A) 78

43. In how many ways can a student choose 5 courses out of 9 if 2 courses are compulsory?

- A) 35
- B) 126
- C) 84
- D) 56

Answer: A) 35

44. Find the number of ways of selecting 9 balls from 6 red, 5 white, and 5 blue balls if each selection consists of 3 balls of each color.

- A) 200
- B) 150
- C) 100
- D) 250

Answer: A) 200

45. How many numbers between 100 and 1000 can be formed with the digits 5, 6, 7, 8, 9 if the repetition of digits is not allowed?

- A) 60
- B) 120
- C) 24
- D) 125

Answer: A) 60

46. In how many ways can 6 persons sit around a rectangular table with 3 seats on each long side?

- A) 720
- B) 360
- C) 120
- D) 24

Answer: A) 720

47. A box contains 7 red and 5 white balls. In how many ways can 4 balls be drawn such that there are 2 red and 2 white balls?

A) 210

B) 350

C) 105

D) 420

Answer: A) 210

48. How many ways can the letters of 'DIRECTOR' be arranged so that the three vowels are never together?

A) 18000

B) 18720

C) 20160

D) 15120

Answer: A) 18000

49. Find the number of ways in which 4 boys and 4 girls can be seated alternately in a row.

A) 1152

B) 576

C) 2304

D) 40320

Answer: A) 1152

50. How many 3-digit numbers can be formed using the digits 0, 2, 4, 6, 8 if repetition is allowed?

A) 100

B) 125

C) 120

D) 75

Answer: A) 100

51. If ${}^{2n}C_3 : {}^nC_2 = 44 : 3$, find n .

A) 6

B) 12

C) 18

D) 24

Answer: A) 6

52. In how many ways can a group of 12 people be divided into 3 groups of 4 people each?

A) 5775

B) 34650

C) 15400

D) 1925

Answer: A) 5775

53. How many words can be formed from the letters of 'TUESDAY' starting with T and ending with Y?

A) 120

B) 24

C) 720

D) 5040

Answer: A) 120

54. There are 15 stations on a railway line. How many different single tickets must be printed to enable a passenger to travel from one station to another?

A) 210

B) 105

C) 225

D) 196

Answer: A) 210

55. In how many ways can 5 distinct rings be worn on 4 fingers of one hand?

A) 1024

B) 625

C) 20

D) 120

Answer: A) 1024

56. The number of subsets of the set $\{1, 2, 3, 4, 5, 6\}$ is:

- A) 64
- B) 32
- C) 128
- D) 12

Answer: A) 64

57. How many ways can 3 men and 3 women sit at a round table such that men and women sit alternately?

- A) 12
- B) 36
- C) 72
- D) 144

Answer: A) 12

58. Find the value of $0! + 1! + 2!$.

- A) 4
- B) 3
- C) 2
- D) 5

Answer: A) 4

59. In how many ways can a jury of 5 be selected from 8 men and 6 women to include exactly 3 men?

- A) 840
- B) 1001
- C) 560
- D) 2002

Answer: A) 840

60. How many diagonals are there in a polygon of 12 sides?

A) 54

B) 66

C) 44

D) 35

Answer: A) 54

61. In how many ways can 6 apples be distributed among 3 boys?

A) 729

B) 216

C) 20

D) 60

Answer: A) 729

62. How many numbers formed with digits 0, 1, 2, 5, 7, 8 will fall between 13 and 1000 if digits can be repeated?

A) 215

B) 214

C) 216

D) 210

Answer: A) 215

63. The number of arrangements of the letters of the word 'CALCUTTA' is:

A) 5040

B) 2520

C) 10080

D) 40320

Answer: A) 5040

64. A tea party is arranged for 16 persons along two sides of a long table with 8 chairs on each side. 4 persons wish to sit on one side and 2 on the other side. In how many ways can they be seated?

A) $10! \times 8!$

B) $(10C4) \times 8! \times 8!$

C) $(10P4) \times 12!$

D) $16!$

Answer: B) $(10C4) \times 8! \times 8!$

65. How many distinct rectangles can be formed on a chessboard?

A) 1296

B) 204

C) 64

D) 1024

Answer: A) 1296

66. If $nP3 = 1320$, then n is:

A) 12

B) 11

C) 10

D) 13

Answer: A) 12

67. In how many ways can 4 letters be put in 4 addressed envelopes so that all letters are in the wrong envelopes (Derangement)?

A) 9

B) 24

C) 23

D) 44

Answer: A) 9

68. How many numbers can be formed using the digits 1, 2, 3, 4, 3, 2, 1 so that the odd digits always occupy the odd places?

A) 18

B) 24

C) 36

D) 12

Answer: A) 18

69. In a group of 15 boys, there are 7 boy-scouts. In how many ways can 12 boys be selected so as to include at least 6 boy-scouts?

A) 175

B) 168

C) 266

D) 250

Answer: C) 266

70. Find the number of ways in which 8 persons can be seated at a round table so that all shall not have the same neighbors in any two arrangements.

A) 2520

B) 5040

C) 1260

D) 40320

Answer: A) 2520

71. How many words can be formed from 'DAUGHTER' such that the word starts with D and ends with R?

A) 720

B) 120

C) 5040

D) 360

Answer: A) 720

72. The number of non-empty subsets of a set having 4 elements is:

A) 15

B) 16

C) 14

D) 12

Answer: A) 15

73. In how many ways can 3 candidates be selected from 8 candidates?

- A) 56
- B) 336
- C) 24
- D) 168

Answer: A) 56

74. If 5 coins are tossed, find the number of possible outcomes.

- A) 32
- B) 25
- C) 10
- D) 16

Answer: A) 32

75. How many 3-digit numbers can be formed using digits 3, 5, 7, 9 with repetition allowed?

- A) 64
- B) 24
- C) 12
- D) 81

Answer: A) 64

76. In how many ways can the letters of the word 'ARTICLE' be arranged so that vowels occupy the even places?

- A) 144
- B) 72
- C) 288
- D) 576

Answer: A) 144

77. How many distinct permutations can be made from the letters of the word 'ASSASSINATION'?

A) 10810800

B) 10800

C) 39916800

D) 1814400

Answer: A) 10810800

78. 6 parallel lines are intersected by 5 other parallel lines. The total number of parallelograms formed is:

A) 150

B) 120

C) 100

D) 90

Answer: A) 150

79. Find the number of ways to arrange 4 boys and 3 girls in a row such that girls are always together.

A) 720

B) 144

C) 576

D) 120

Answer: A) 720

80. A code consists of 2 distinct English letters followed by 2 distinct non-zero digits. How many such codes are possible?

A) 46800

B) 65000

C) 58500

D) 48600

Answer: A) 46800

81. In how many ways can 5 boys and 3 girls be arranged in a row so that the girls are never together?

- A) 14400
- B) 2400
- C) 7200
- D) 3600

Answer: A) 14400

82. How many numbers greater than 5000 can be formed with the digits 3, 4, 5, 6, 7 without repetition?

- A) 72
- B) 120
- C) 96
- D) 48

Answer: A) 72

83. The number of ways in which a committee of 6 members can be formed from 8 gentlemen and 4 ladies so that the committee contains at least 3 ladies is:

- A) 252
- B) 672
- C) 420
- D) 336

Answer: A) 252

84. How many ways can 4 prizes be given to 3 boys if each boy can receive at most one prize?

- A) 0 (Not possible)
- B) 12
- C) 24
- D) 81

Answer: A) 0 (Not possible as prizes > boys in 1-1 mapping, assuming distinct prizes and 1 per boy. If 1 boy can get multiple, it's $3^4 = 81$. Let's assume the question meant 3 prizes to 4 boys at most 1, or just distribution. Re-evaluating: 'Given to 3 boys' usually implies 3^4 . 'At most one' implies P. But 4 items into 3 slots 1 each is impossible. Let's assume standard distribution $3^4 = 81$.)

Correction: If the question is "How many ways can 3 prizes be given to 4 boys if each boy receives at most 1 prize?", the answer is $4P3 = 24$. Let's fix the question to valid solvable logic.

Revised Question 84: In how many ways can 3 prizes be distributed among 4 boys if each boy can get at most one prize?

Answer: 24

84. In how many ways can 3 distinct prizes be distributed among 4 boys if each boy can get at most one prize?

A) 24

B) 64

C) 12

D) 81

Answer: A) 24

85. In how many ways can 5 differently colored beads be arranged in a necklace?

A) 12

B) 24

C) 60

D) 120

Answer: A) 12

86. Find n if ${}^{2n}P_3 = 100$. (Correction: ${}^{2n}P_3$ implies $(2n)(2n-1)(2n-2)$. Let's use simple numbers).

Question: If ${}^nC_3 = 10$, find n .

A) 5

B) 6

C) 4

D) 3

Answer: A) 5

87. How many 3-digit numbers can be formed using digits 1, 3, 5, 7, 9 divisible by 5 (no repetition)?

A) 12

B) 20

C) 24

D) 16

Answer: A) 12

88. A polygon has 9 diagonals. How many sides does it have?

A) 6

B) 5

C) 7

D) 8

Answer: A) 6

89. A student has 6 pants and 4 shirts. In how many ways can he wear a dress?

A) 24

B) 10

C) 36

D) 12

Answer: A) 24

90. In how many ways can 3 distinct numbers be chosen from the set $\{1, 2, 3, \dots, 9\}$ such that they are in Arithmetic Progression?

A) 16

B) 12

C) 14

D) 18

Answer: A) 16

91. How many words can be formed by rearranging the letters of the word 'PENCIL' so that N is always next to E?

A) 240

B) 120

C) 720

D) 480

Answer: A) 240

92. In a shelf, there are 5 English and 4 Hindi books. In how many ways can they be arranged so that all Hindi books are together?

- A) 17280
- B) 2880
- C) 1440
- D) 3600

Answer: A) 17280

93. Find the number of ways to distribute 12 identical mangoes among 3 children.

- A) 91
- B) 78
- C) 66
- D) 55

Answer: A) 91

94. 4 boys and 2 girls are to be seated in a row. Find the probability that the two girls sit together. (Or simply number of ways).

Number of ways 4 boys and 2 girls sit so girls are together:

- A) 240
- B) 120
- C) 480
- D) 720

Answer: A) 240

95. How many 6-digit numbers can be formed using the digits 0, 1, 3, 5, 7, 9 which are divisible by 10 and no digit is repeated?

- A) 120
- B) 720
- C) 24
- D) 60

Answer: A) 120

96. If ${}^{15}C_r = {}^{15}C_{(r-1)}$, find r .

A) 8

B) 7

C) 9

D) 6

Answer: A) 8

97. The number of ways of dividing 15 men and 15 women into 15 couples (each consisting of a man and a woman) is:

A) 15!

B) 15

C) 225

D) 30!

Answer: A) 15!

98. A box contains 2 white, 3 black, and 4 red balls. In how many ways can 3 balls be drawn such that there is at least one black ball?

A) 64

B) 84

C) 20

D) 50

Answer: A) 64

99. In how many ways can 6 teachers be assigned to 3 sections of a class?

A) 729

B) 216

C) 18

D) 120

Answer: A) 729

100. Find the number of ways in which the letters of the word 'MONDAY' can be written such that the word does not begin with M.

A) 600

B) 720

C) 120

D) 480

Answer: A) 600